

ASSIGNMENT: 01

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Subject: Computer Organization and Assembly Language

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NUMBER SYSTEM CONVERSIONS

Q1. Convert (157)₈ into hexadecimal.

Solution:

Step 1: Convert octal to decimal.

Using the place values of 8 from right to left:

$$(157)_8 = 1 \times 8^2 + 5 \times 8^1 + 7 \times 8^0$$

$$= 1 \times 64 + 5 \times 8 + 7 \times 1$$

$$= 64 + 40 + 7 = 11110$$

Step 2: Convert 11110 into hexadecimal.

$$111 / 16 = 6, \text{ remainder } 15 (F)$$

$$6 / 16 = 0, \text{ remainder } 6$$

Reading remainders from bottom to top: 6F

Answer: (157)₈ = (6F)₁₆

Q1. Convert (6F)₁₆ into octal.

Solution:

Step 1: Convert hexadecimal to decimal.

Here F = 15.

$$(6F)_{16} = 6 \times 16^1 + 15 \times 16^0$$

$$= 6 \times 16 + 15 \times 1 = 96 + 15 = 11110$$

Step 2: Convert 11110 into octal.

$$111 / 8 = 13, \text{ remainder } 7$$

$$13 / 8 = 1, \text{ remainder } 5$$

$$1 / 8 = 0, \text{ remainder } 1$$

Reading remainders from bottom to top: 157

Answer: (6F)₁₆ = (157)₈

Q2. Convert (156)₁₀ into octal.

Solution:

Step 1: Convert decimal to octal by repeated division by 8.

$$156 / 8 = 19, \text{ remainder } 4$$

$$19 / 8 = 2, \text{ remainder } 3$$

$$2 / 8 = 0, \text{ remainder } 2$$

Reading remainders from bottom to top: 234

Answer: (156)₁₀ = (234)₈

Q2. Convert (234)₈ into decimal.

Solution:

Step 1: Convert octal to decimal using place values.

$$\begin{aligned}
 (234)_8 &= 2 \times 8^2 + 3 \times 8^1 + 4 \times 8^0 \\
 &= 2 \times 64 + 3 \times 8 + 4 \times 1 \\
 &= 128 + 24 + 4 = 156_{10}
 \end{aligned}$$

Answer: $(234)_8 = (156)_{10}$

Q3. Convert $(2A)_{16}$ into binary.

Solution:

Step 1: Convert hexadecimal to decimal.

Here A = 10.

$$\begin{aligned}
 (2A)_{16} &= 2 \times 16^1 + 10 \times 16^0 \\
 &= 2 \times 16 + 10 = 42
 \end{aligned}$$

Step 2: Convert 4210 into binary.

$$42 / 2 = 21, \text{ remainder } 0$$

$$21 / 2 = 10, \text{ remainder } 1$$

$$10 / 2 = 5, \text{ remainder } 0$$

$$5 / 2 = 2, \text{ remainder } 1$$

$$2 / 2 = 1, \text{ remainder } 0$$

$$1 / 2 = 0, \text{ remainder } 1$$

Reading remainders from bottom to top: 101010

Answer: $(2A)_{16} = (101010)_2$

Q3. Convert $(101010)_2$ into hexadecimal.

Solution:

Step 1: Convert binary to decimal.

$$\begin{aligned}
 (101010)_2 &= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\
 &= 32 + 0 + 8 + 0 + 2 + 0 = 42
 \end{aligned}$$

Step 2: Convert 4210 into hexadecimal.

$$42 / 16 = 2, \text{ remainder } 10 \text{ (A)}$$

$$2 / 16 = 0, \text{ remainder } 2$$

Reading remainders from bottom to top: 2A

Answer: $(101010)_2 = (2A)_{16}$

Q4. Convert $(101101)_2$ into decimal.

Solution:

Step 1: Use the powers of 2 from right to left.

$$1 \times 2^5 = 32$$

$$0 \times 2^4 = 0$$

$$1 \times 2^3 = 8$$

$$1 \times 2^2 = 4$$

$$0 \times 2^1 = 0$$

$$1 \times 2^0 = 1$$

Adding: $32 + 0 + 8 + 4 + 0 + 1 = 45$

Answer: $(101101)_2 = (45)_{10}$

Q4. Convert $(45)_{10}$ into binary.

Solution:

Step 1: Convert decimal to binary by repeated division by 2.

$45 / 2 = 22$, remainder 1

$22 / 2 = 11$, remainder 0

$11 / 2 = 5$, remainder 1

$5 / 2 = 2$, remainder 1

$2 / 2 = 1$, remainder 0

$1 / 2 = 0$, remainder 1

Reading remainders from bottom to top: 101101

Answer: $(45)_{10} = (101101)_2$

MEMORY SPACE

Q5. Find the memory space required by a String.

Example: String = "Mariam"

Number of characters = 6

Each character = 1 byte

Memory required = $6 \times 1 = 6$ bytes

Answer: 6 bytes

Q6. Find the memory space required by an Integer (int).

Assume 1 int = 4 bytes.

Memory required = 4 bytes

Answer: 4 bytes

Q7. Find the memory space required by a Double.

Assume 1 double = 8 bytes.

Memory required = 8 bytes

Answer: 8 bytes

Q8. Find the memory space required by a One-Dimensional Array.

Example: int A[5]

Number of elements = 5

Size of each int = 4 bytes

Memory = $5 \times 4 = 20$ bytes

Answer: 20 bytes

Q9. Find the memory space required by a Two-Dimensional Array.

Example: int A[3][4]

Rows = 3

Columns = 4

Total elements = $3 \times 4 = 12$

Size of each int = 4 bytes

Memory = $12 \times 4 = 48$ bytes

Answer: 48 bytes